# MM-RCNN: Toward Few-Shot Object Detection in Remote Sensing Images With Meta Memory

Jianxiang Li, Yan Tian, Yiping Xu, Xinli Hu, Zili Zhang, Hu Wang, and Yiwen Xiao

***Abstract*— In the area of optical remote sensing image processing, object detection is among the utmost essential and difficult tasks. By virtue of the excellent feature representation abilities of deep convolutional neural networks (DCNNs), the performance of remote sensing object detection has recently increased dramatically. However, DCNN-based methods necessitate sufficient labeled training samples and tend to experience a considerable performance fall when training examples are insufficient. Many of the recently developed few-shot object detection (FSOD) methods attempt to solve the issue using the idea of meta-learning, which intends to extract knowledge that can be generalized across various tasks. Despite its success, the negligence of the knowledge learned in the past and interclass correlations hinder the detection ability of novel classes. In this article, we propose a new meta-memory-based FSOD approach named MM-RCNN. Specifically, MM-RCNN adopts a memory module to store each category's knowledge learned in the training stage and the memory-based external attention (MEA) to aggregate all the categories' information simultaneously. Based on MEA, we design two feature enhancement modules for the region proposal network (RPN) and detection head to boost the performance. Experiments over two remote sensing benchmarks, i.e., DIOR and NWPU VHR10, verify the capability of our method (0.239 and 0.557 average mAP across all settings).**

***Index Terms*— Attention mechanism, few-shot object detection (FSOD), remote sensing images.**

## I. Introduction

OBJECT detection is among the most essential and challenging tasks in the area of optical remote sensing image (RSI) processing [1], [2], [3], [4], which is defined as locating and recognizing the target objects of interest in the input image. It has enabled various real-world applications such as urban planning, change detection, and land surveying.

In the past decades, numerous approaches [1], [2] have been explored to detect various geospatial targets [5], [6], [7], [8], [9] (e.g., vehicles, ships, and roads) in RSIs. The pipeline of the traditional remote sensing object detectors could generally be divided into three steps [1]: 1) proposal (or regions of interest, RoIs) generation; 2) the extraction of feature vector encoded by hand-crafted low-level visual descriptors; and 3) region classification with a classifier. The performance of these methods is highly influenced by the hand-crafted features.

Due to the advancement of deep convolutional neural network (DCNN) [10] and publicly available large-scale benchmarks [11], [12], the DCNN-based object detection approaches [13], [14], [15], [16], [17] have made considerable progress in natural scene images due to the excellent feature representation capability. In the remote sensing community, current studies [18], [19], [20] build their approaches upon those prevalent DCNN-based architectures, such as Faster RCNN [13] and YOLO series [14], [15], [16].

Despite the excellent performance that the DCNN-based approaches have achieved so far, they have the same flaw [21], [22], [23]: training a DCNN model necessitates a large-scale, heterogeneous dataset. However, the annotation of a large number of samples needs massive human labor, and it is sometimes unavailable for rare categories. These models experience severe overfitting and also have poor generalization capability due to insufficient and severely imbalanced training data [23]. In contrast, humans can easily recognize new objects with only a few samples.

Recent advancements [24], [25], [26], [27] in few-shot object detection (FSOD) for natural scenes have shown great success in learning to locate and recognize novel class objects with limited (few-shot) data samples. The existing FSOD methods in natural scene images could be approximately divided into three groups: 1) data-augmentation-based approaches [28], [29]; 2) transfer learning or fine-tuning-based approaches [26], [27]; and 3) meta-learning-based approaches [24], [25]. Data augmentation, which serves as a preprocessing step, tries to inflate the training dataset to make it large enough to provide reliable hypotheses. However, expanding training samples only improves the model's anti-interference performance but not the data distribution. The transfer learning methods entail reusing network parameters that have been pretrained on a base domain dataset with abundant samples to increase the generalization performance on a novel domain dataset with insufficient examples. Nevertheless, the transfer learning methods tend to neglect the severe domain shift problem. Recently, meta-learning approaches have become popular in FSOD. They allow for quick adaptation of novel classes, which is particularly beneficial for practical cases. However, they usually involve a complicated episodic training scheme and data organization.

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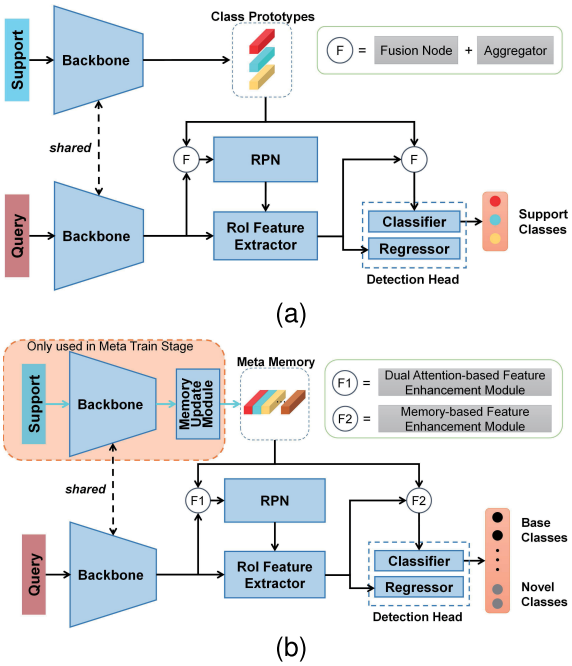



Fig. 1. Comparisons of the prior meta-learning-based FSOD approaches and our meta-memory-based FSOD approach. (a) Prior meta-learning-based FSOD approaches with two branches only detect the objects belonging to the support classes, which neglect the interclass correlation. Besides, they obtain the class prototypes through current support images and do not fully use the whole support set. (b) Proposed FSOD approach uses a meta memory module to store the information of all the classes, which is only updated in the meta training stage. Moreover, it uses meta memory as external guidance to capture the interclass relationship.

Despite the great success of the meta-learning-based methods achieved in FSOD, there are two underlying disadvantages that hinder better exploitation of the learned knowledge, as illustrated in Fig. 1(a). First, the current meta-learning-based FSOD methods neglect a critical problem [30]: how previous information could be used for the new task when the episodes are trained one by one. We argue that a few-shot learning model should identify the relationship between episodes and make full use of previously learned knowledge. Second, most meta-learning-based FSOD methods adopt feature reweighting [25] or its variants to explore the relationship between support and query images, which only deal with one support class at a time to detect all-class objects. We argue that instead of seeing one class at a time with repeated runs, the object detector should see various classes together at the same time so that it can better learn to distinguish similar classes.

To overcome these difficulties mentioned above, a novel meta-learning-based FSOD approach named meta-memory-based RCNN (MM-RCNN) is proposed in this article, which is built upon the widely used object detection framework Faster RCNN [13]. Specifically, we adopt the memory mechanism to deal with the inadequate use of the previous knowledge problem, as shown in Fig. 1(b). The meta memory module stores the meta knowledge of each category, which can be updated in the meta training stage. To alleviate the second problem of overlooking the interclass correlations, we propose the memory-based external attention (MEA), which takes the meta memory as external guidance. The MEA aggregates all the categories' information simultaneously to reduce false classification via the capture of interclass correlations.

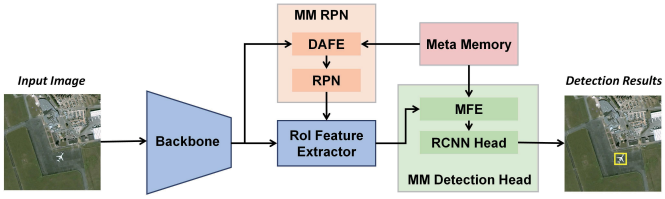



Fig. 2. General architecture of our method MM-RCNN in the inference stage. It consists of four modules: the backbone for feature extraction, a meta memory module that stores meta information of all the classes, a modified RPN with the DAFE module to generate candidate regions, and the modified detection head with the MFE module to obtain the object's category and bounding box.

As illustrated in Fig. 2, MM-RCNN mainly has three novel modules: a meta memory module, a dual attention-based feature enhancement (DAFE) module, and the memory-based feature enhancement (MFE) module. The meta memory module stores the meta knowledge of each category and serves as external guidance for DAFE and MFE. We adopt a momentum update strategy to refine the memory unit in the meta training stage iteratively. The DAFE and MFE are designed based on the MEA. The distinction is that DAFE has two attention mechanisms to enhance the feature for RPN, but MFE directly uses MEA to enhance the proposal features. The DAFE module first applies self-attention to mine the relationships between pixels of the query feature map, and then uses modified MEA to filter out the features that do not match any class. The MFE module uses the meta memory to enhance the proposal features, further improving the classification performance. As with other meta-learning-based FSOD approaches [24], the proposed method needs two training stages: a base training stage to obtain a basic detector by learning a supervised representation with abundant training data and a meta training stage to learn the meta knowledge of each class and obtain an FSOD detector, as shown in Fig. 3. After meta training, the information of all the categories is kept in the meta memory module. Hence, our method does not need the support branch in the meta testing stage, as shown in Fig. 1(b).

To summarize, our work has three major contributions.

1) We propose a meta-learning-based MM-RCNN framework designed for the FSOD task in remote sensing scene. We adopt a memory module that is kept during episodic training to maintain a stable and consistent knowledge, which learns an effective prototype representation for each class.
2) We propose a novel MEA, which can aggregate all the categories' information simultaneously to capture interclass correlations.
3) We apply MEA to the DAFE module and MFE module, which are proposed to enhance the feature maps of RPN and RoI features, respectively.

The rest of this article is organized as follows. Section II gives a brief overview of the related work. In Section III,

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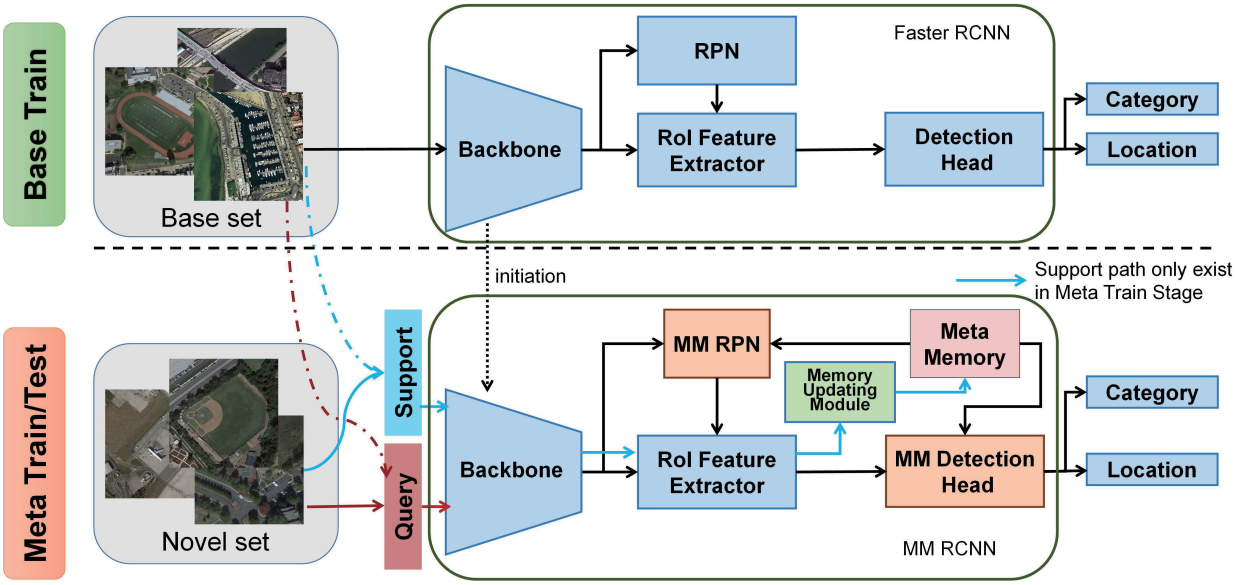



Fig. 3. Training framework of the proposed method. First, we train the vanilla Faster RCNN using the base set with abundant training samples. Then we adopt the episodic training strategy with the union of the base set and novel set in the meta training stage. In this stage, the proposed meta memory module is refined via the memory updating module, and the DAFE and MFE modules are trained. In the meta testing stage, MM-RCNN uses meta memory to detect all the classes in the query image.

we describe the preliminary knowledge about MM-RCNN. Then, we introduce our MM-RCNN in detail in Section IV. Finally, we perform comprehensive experiments in Section V and conclude the work in Section VI.

## II. Related Work

In this section, the related works are reviewed from four perspectives: object detection in natural scenes, object detection in remote sensing scenes, FSOD, and the attention mechanism in computer vision.

### A. Object Detection in Natural Scenes

Object detection is among the utmost basic yet challenging computer vision tasks. In recent years, DCNN-based object detectors have achieved remarkable progress due to their powerful learning capacity. Most DCNN-based detectors can be categorized into proposal-based (two-stage) and proposal-free (one-stage) object detection approaches. The proposal-based approaches divide the detection task into two steps: generating region proposals and refining these proposals. For example, Faster RCNN [13] first uses a region proposal network (RPN) to obtain several candidate regions of interest, and then categorizes and regresses these candidate regions to obtain final detection results. The proposal-based methods have achieved promising performance while suffering the low inference speed. On the contrary, proposal-free object detectors can obtain the detection results via one-step inference, such as YOLO series [14], [15], [16] and RetinaNet [17]. Proposal-free detectors are efficient, but their accuracy is inferior to the proposal-based detectors. To eliminate the need for numerous predefined anchor boxes, anchor-free approaches, such as CornerNet [31], ExtremeNet [32], and CenterNet [33], have received much attention recently. Anchor-free detectors directly regress the key points, such as corner points [31], extreme points [32], or one center point [33], and then group them into bounding boxes. Recently, the transformer-based object detection methods [34], [35], [36] have become popular for the potential to eliminate postprocessing by the one-to-one assignment strategy. For example, DETR [34] directly outputs the predictions without the widely used nonmaximum suppression. However, DETR needs more training time, and the performance is unsatisfactory. To alleviate the problems faced by DETR, several methods are proposed to speed up the convergence speed [35] or improve the design of the encoder and decoder in DETR [36].

### B. Object Detection in Remote Sensing Scenes

The rapid progress of the DCNN-based object detection algorithms mentioned above has inspired a lot of interest in object detection research in remote sensing scenes. Note that the targets of interest in RSIs usually have large variations in scales, aspect ratios, and orientations. Many object detection approaches are developed to solve these problems. For example, Cheng et al. [18] presented RIFD-CNN to explicitly address the problems related to object orientation, intraclass diversity, and interclass similarity. Wu et al. [19] proposed ORSIm detector to handle objective scaling and rotation problem. By combining rotation-invariant channel features extracted from the frequency domain with shift-invariant channel features extracted from the original spatial domain, they introduced a novel spatial-frequency channel feature. To tackle the rotation variation and visual ambiguity problems, Li et al. [20] proposed rotation-insensitive RPN by incorporating additional multioriented anchors into the original RPN and a feature fusion network using on local contextual information. Huang et al. [37] proposed the nonlocal-aware

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pyramid attention to capture the multiscale nonlocal dependencies and multioriented edge features. Deng et al. [38] designed a multiscale object proposal network (MS-OPN) to detect objects with large size variability on both the optical and synthetic aperture radar (SAR) image datasets. Pang et al. [39] designed a self-reinforced and unified network to detect very small objects from complex backgrounds. Li et al. [40] proposed to detect small vehicles by enhancing the object features through a target-guided feature super-resolution module.

Besides, there are some methods [41], [42], [43], [44] aiming at the oriented object detection task. For example, Ding et al. [43] proposed RoI Transformer with a spatial transformation module to transform horizontal proposals into rotated proposals. Xu et al. [44] proposed to slide four vertexes on the respective sides of the minimum bounding rectangle to represent the oriented bounding box. Shi et al. [42] proposed the VCSOP detector to detect arbitrarily oriented vehicles in an anchor-free manner. Huang et al. [45] proposed a Gaussian object label assignment strategy, which adaptively reflected objects' shape and orientation. LO-Det [46] adopted a channel separation–aggregation structure to achieve low complexity for the sake of edge computing on embedded devices. Moreover, most RSI processing and analysis approaches are merely designed for single modalities, such as RGB [44], multispectral or hyperspectral [47], and SAR [38], which leads to limited ability for the lack of diverse information. Recently, with the open availability of multimodel RS data, multimodality learning (MML) [48], [49] has been proposed to explore various properties at a more detailed level. Wu et al. [50] proposed the Ms-AFt framework to efficiently explore helpful information from multisource data, namely, RGB and digital surface models (DSM) images, to label and detect vehicles in the RSIs automatically.

### C. Few Shot Object Detection

In recent years, FSOD has become popular for its ability to detect novel classes' objects with the help of limited labeled samples.

Currently, the approaches for FSOD can be categorized into three groups: 1) data-augmentation-based approaches [28], [29], [51]; 2) fine-tuning-based approaches [26], [27]; and 3) meta-learning-based approaches [24], [25]. The data-augmentation-based methods try to supplement training samples or semantic knowledge for novel categories. For example, Li and Li [28] and Zhang and Wang [29] proposed to expand the support set using specialized data augmentation techniques. Zhu et al. [51] proposed using external semantic knowledge to boost the classification performance in FSOD. The fine-tuning-based methods aim at transferring the knowledge acquired from the base classes to the novel classes, which are simple yet effective. Wang et al. [27] proposed a two-stage fine-tune-based approach (TFA) which freezes all the learned parameters of backbone and RPN but only fine-tunes detection heads with novel class. FSCE [26] improves TFA by introducing a contrastive head and fine-tuning more detector modules. The meta-learning-based approaches introduce a meta-learner to assist the model detect novel classes by learning class-level meta knowledge. Kang et al. [25] introduced a simple module called feature reweighting, which reweights the query features with support prototypes, to YOLOv2 [15]. Yan et al. [24] proposed Meta-RCNN with a meta-learner predictor head to endow the traditional Faster/Mask RCNN [13], [52] on the FSOD/segmentation tasks.

All these methods mentioned above are developed for natural images. In contrast, relatively few pioneering works aim at the FSOD task in remote sensing scenes. Li et al. [53] presented a meta-learning-based approach named FSODM, the first method for FSOD in remote sensing images. FSODM was designed in a multiscale architecture based on feature reweighting [25]. Based on Meta RCNN, Cheng et al. [54] proposed P-CNN with a prototype-guided RPN to provide class-aware clues for high-quality candidate proposal generation. Inspired by TFA [27], Zhao et al. [55] proposed a fine-tuning-based method PAMS-Det by introducing an involution-based backbone and a path-aggregation module (PAM) to enhance localization performance.

### D. Attention Mechanism in Computer Vision

In the past decade, the attention mechanism, which is designed to reallocate pixel values according to the activation, has been an active research topic in computer vision. According to the data domain, the attention mechanism can be roughly categorized into channel attention [56], spatial attention [57], and hybrid category [58]. SENet [56] captures the channelwise relationships of the feature maps and then rescales each channel feature through the squeeze-and-excitation block. Dai et al. [57] proposed deformable convolution and deformable RoI pooling to adaptively choose important regions, which is desirable for the object detection task. CBAM [58] stacks the channel attention module and spatial attention module in series to combine their advantages.

Recently, self-attention [59] and its variants [60], [61] are among the most successful cases and shows great potential in various vision tasks. Wang et al. [60] proposed the nonlocal blocks to capture long-range dependencies, which promoted the performance of object detection. Huang et al. [61] proposed criss-cross attention to obtain dense contextual information with less computational and memory cost. However, these methods follow self-attention to mine the relationships between pixels within the input data sample and ignore the correlations between pixels within other samples. Then, Guo et al. [62] proposed external attention to mine the correlations between the input and all other samples.

## III. Preliminary Knowledge

### A. Problem Formulation

As in various previous work [25], [27], we adopt the following settings for the FSOD task in our work. There are two training sets, i.e., a base set $D_{\text{base}}$ with abundant annotated instances for each class $c \in C_{\text{base}}$ and a novel set $D_{\text{novel}}$ with insufficient annotated samples per class $c \in C_{\text{novel}}$. It is worth noting that the base classes $C_{\text{base}}$ and the novel classes $C_{\text{novel}}$ are two disjoint sets, namely, $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. For each

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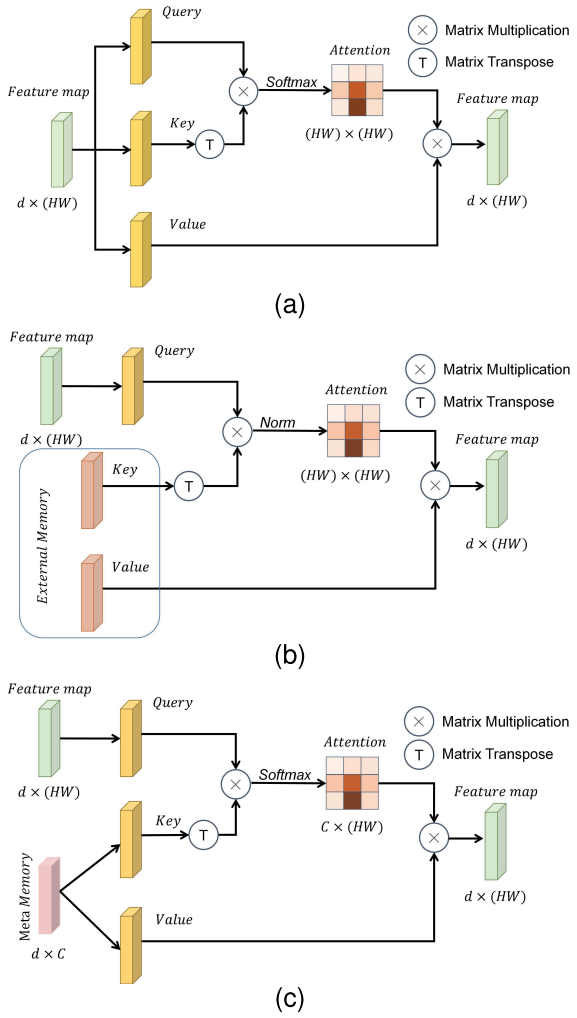



Fig. 4. Comparisons of self-attention, external attention, and the proposed MEA. (a) Self-attention mechanism. (b) External attention mechanism. (c) Proposed MEA mechanism.

sample $(x, y) \in D_{\text{base}} \cup D_{\text{novel}}$, $x$ is an input image containing $N$ labeled instances, and $y = \{(c_i, b_i), i = 1, \ldots, N\}$ represents the annotations with class $c_i \in C_{\text{base}} \cup C_{\text{novel}}$ along with the bounding box locations $b_i$ in the image $x$. Note that the number of labeled samples per class $c \in C_{\text{novel}}$ is called shots.

In general, FSOD aims to improve the detection performance for novel classes while maintaining the detection performance for base classes.

### B. Revisiting Self-Attention and External-Attention

Self-attention [59] is a useful feature enhancement technique that refines an input feature using a linear combination of self-values. As illustrated in Fig. 4(a), considering a feature map $\mathbf{F} \in \mathbb{R}^{d \times N}$, where $d$ denotes the channel numbers and $N = H \times W$ represents the total pixels, self-attention first applies three linear layers or $1 \times 1$ convolutional layers on $\mathbf{F}$ and get three matrices: the query $\mathbf{Q} \in \mathbb{R}^{d' \times N}$, the key $\mathbf{K} \in \mathbb{R}^{d' \times N}$, and the value $\mathbf{V} \in \mathbb{R}^{d \times N}$. Note that $d'$ is usually set to be less than $d$ for information bottleneck. Next,

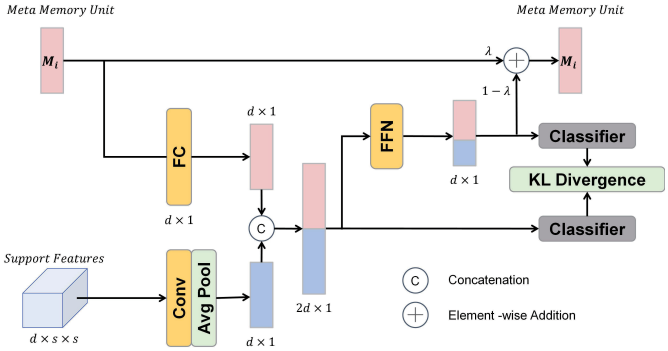



Fig. 5. Meta memory unit updating scheme.

self-attention is computed as follows:

$$\mathbf{A} = \text{softmax}\left(\mathbf{Q}^T \mathbf{K}\right) \tag{1}$$

$$\mathbf{F}_{\text{out}} = \mathbf{AV} \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ indicates the attention matrix and $\mathbf{F}_{\text{out}}$ represents the output of the self-attention module.

Since self-attention only mines the relationships between pixels within a data sample, it ignores potential correlations between pixels in different samples, which could limit its ability and adaptability.

Thus, Guo et al. [62] proposed to calculate the external attention between the query $\mathbf{Q}$ and an external key $\mathbf{M}_k \in \mathbb{R}^{d' \times N}$, as illustrated in Fig. 4(b), via

$$\mathbf{A} = \text{norm}\left(\mathbf{Q}^T \mathbf{M}_K\right) \tag{3}$$

$$\mathbf{F}_{\text{out}} = \mathbf{AM}_V \tag{4}$$

where $\mathbf{M}_V \in \mathbb{R}^{d \times N}$ is another external unit, and norm$(\cdot)$ is a double-normalization process that consists of two normalizations separately applied on rows and columns.

In our work, instead of using two memory units which are regarded as dictionaries for the whole dataset in the original external attention, we adopt the meta memory as the external guidance and propose MEA, which is used in the DAFE and MFE modules, as shown in Fig. 4(c).

## IV. Proposed Method

### A. Overview Framework

As shown in Fig. 1(b), our MM-RCNN mainly has three novel components, i.e., a meta memory updating module for class-aware knowledge refining, DAFE for the query features' enhancement, and MFE for the proposal features' enhancement.

### B. Meta Memory Updating Module

Since the targets of interest in RSIs usually have large variances in aspect ratios and orientations [43], [44], it is possible that the objects belonging to the same category appear in different directions. Therefore, directly taking the average features of K objects from the same class as its prototype may lead to confusion.

Meta knowledge has been proven helpful in few-shot learning, and some recent advanced methods [30] have adopted a

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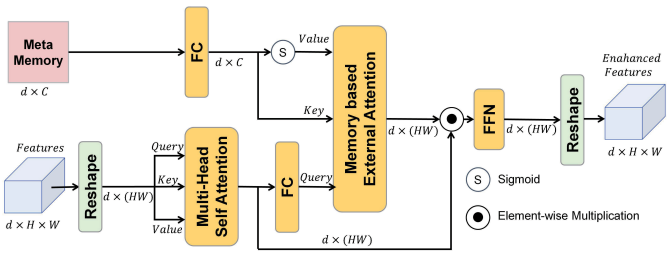



Fig. 6. Architecture of the DAFE module.

memory mechanism to store the meta knowledge. In this work, we adopt a memory bank (meta memory), which holds the meta knowledge of each class, to use the knowledge learned in the past fully.

The meta memory is denoted as $\mathbf{M} \in \mathbb{R}^{d\times C}$, where $C$ indicates the total number of classes and $d$ indicates the channels of embedding features. $\mathbf{M}$ is initialized with one support sample's feature extracted with the model trained in the base training stage. Then refining of the memory unit is conducted iteratively to aggregate discriminative information and dilute object-irrelevant nuisances in the meta training stage.

As shown in Fig. 5, to progressively purify the meta memory, we first extract the RoI features of the support sample and then conduct global average pooling to obtain the support vector $f^l_{i,\text{support}} \in \mathbb{R}^{d\times 1}$ at the episode of $l$. Then we concatenate the support vector $f^l_{i,\text{support}}$ with the memory unit $\mathbf{M}_i \in \mathbb{R}^{d\times 1}$ that belongs to the same class. We pass the concatenation $f^l_{i,c} \in \mathbb{R}^{2d\times 1}$ to a fully connected feed-forward network (FFN) and refine the memory unit with the output of FFN. We adopt a momentum update strategy to refine the $i$th category memory $\mathbf{M}_i$, as

$$\mathbf{M}^{l+1}_i = \lambda \mathbf{M}^l_i + (1-\lambda) f^l_{i,p} \tag{5}$$

where $\lambda \in [0, 1)$ is a momentum coefficient and $f^l_{i,p}$ denotes the output of the FFN at the episode of $l$. To ensure the FFN purifies the discriminative information, we adopt the constraint proposed by Liu et al. [30], as

$$\mathcal{L}_r = \mathbb{E}_{f^l_{i,c}}\left[\mathbb{E}_{z\sim\phi\left(z\middle|f^l_{i,c}\right)}\left[D_{kl}\left[p\left(y\middle|f^l_{i,c}\right), p\left(y\middle|f^l_{i,p}\right)\right]\right]\right] \tag{6}$$

where $\phi$ denotes the parameters of the FFN, $D_{kl}[\cdot,\cdot]$ represents the KL-divergence, and $y$ indicates the label of the support sample. $p(y|\cdot)$ is the output of the classifier and denotes conditional distribution. $\mathbb{E}$ represents the expectation.

### C. DAFE Module

Prior meta-learning-based FSOD methods usually adopt feature reweighting [53] or its variants [54] to aggregate the query features and support feature to obtain their relationship. However, they ignore the interclass correlation, which is important for multiclass recognition. In contrast, we propose to apply two attention mechanisms in DAFE to aggregate query features with the meta memory for the class-agnostic RPN. The DAFE module first applies self-attention to find similar pixels in the query feature map, and then uses MEA to filter out the features that do not match any class.

The MEA can aggregate all the classes simultaneously to capture interclass correlation and further reduce false classification. As shown in Fig. 6, given the query feature $\mathbf{F} \in \mathbb{R}^{d\times H\times W}$, we first reshape it to $\mathbf{F}' \in \mathbb{R}^{d\times HW}$ and use multihead self-attention to enhance it. Then we conduct feature matching via single-head external attention. The matching coefficients are calculated as

$$\mathbf{A} = \text{softmax}\left[\frac{(\mathbf{F}'\mathbf{W}_F)(\mathbf{M}\mathbf{W}_M)^T}{\sqrt{d}}\right] \tag{7}$$

where $\mathbf{W}_F$ and $\mathbf{W}_M$ are used to project $\mathbf{F}'$ and $\mathbf{M}$ into the same embedding space. Subsequently, the output of the MEA can be produced by

$$\mathbf{F}_{\text{out}} = \mathbf{A}\sigma(\mathbf{M}) \odot \mathbf{F} \tag{8}$$

where $\sigma(\cdot)$ and $\odot$ denote the sigmoid function and Hadamard product, respectively. $\sigma(\mathbf{M})$ serves as class-related feature filters for each class. By applying $\mathbf{A}$ to $\sigma(\mathbf{M})$, the features not matched to any class can be filtered. Then, we feed the filtered features into the original RPN of the Faster RCNN to get region proposals.

### D. MFE Module

For the purpose of improving the generalization of detectors, we explore using meta memory to enhance RoI features, as shown in Fig. 7. We use the RoI Align [52] operation to get RoI features from proposals. The proposal feature is denoted as $\mathbf{P} \in \mathbb{R}^{n\times d\times s\times s}$, where $n$, $d$, and $s$ indicate the number of proposals, the channel numbers of features, and the size of proposals, respectively.

For proposal features $\mathbf{P}$ and memory $\mathbf{M}$, we separately apply a $1 \times 1$ convolutional layer and a fully connected layer to project $\mathbf{P}$ and $\mathbf{M}$ into the same embedding space to obtain $\mathbf{P}'$ and $\mathbf{M}'$. Then, we calculate the soft attention $\mathbf{A}_k$ and obtain the enhanced proposal features $\mathbf{H}_k$, via

$$\mathbf{A}_k = \text{softmax}\left(\mathbf{P}'_k \mathbf{M}'^{\mathrm{T}}\right) \tag{9}$$

$$\mathbf{H}_k = f\left(\text{concat}\left(\mathbf{P}'_k, \mathbf{A}_k\mathbf{M}'\right); \theta\right) + \mathbf{P}_k \tag{10}$$

where $\mathbf{P}'_k \in \mathbb{R}^{s^2\times m}$ represents the $k$th proposal of $\mathbf{P}'$, and $k = 1, \ldots, n$. $\mathbf{A}_k \in \mathbb{R}^{s^2\times C}$ denotes the attention weights. $f(\cdot;\theta)$ consists of two convolutional layers with the ReLU activation function. $\mathbf{P}_k \in \mathbb{R}^{s\times s\times m}$ indicates the $k$th proposal of $\mathbf{P}$. Next, we get the enhanced features $\mathbf{H} \in \mathbb{R}^{n\times s\times s\times d}$, which aggregates the information of meta memory. Finally, we feed the enhanced features into the detection head to get the final classes and locations.

### E. Model Training Framework

As other meta-learning-based FSOD methods like Meta-RCNN [24], our approach can be summarized into three stages, i.e., base train, meta train, and meta test, as illustrated in Fig. 3.

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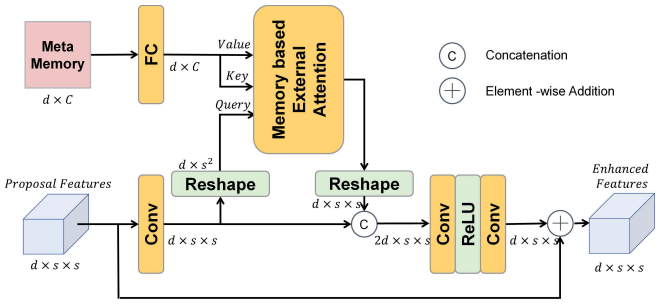



Fig. 7. Architecture of the MFE module.

TABLE I
DIFFERENT SPLIT SETTINGS ON THE DIOR DATASET

| # Split | Novel classes | | | | | Base classes |
|---|---|---|---|---|---|---|
| 1 | BF | BC | BR | CH | SH | rest |
| 2 | APL | APO | ETS | HA | GTF | rest |
| 3 | DAM | GF | STO | TC | VE | rest |
| 4 | ESA | OP | STA | TS | WM | rest |

*Phase I (Base Train):* Following most FSOD methods, we first train the vanilla Faster RCNN model with the backbone pretrained on ImageNet [12] using the base set $D_{\text{base}}$ with abundant training samples to learn a supervised representation. The loss function is defined as

$$\mathcal{L}_{\text{FRCNN}} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{rcnn}} \tag{11}$$

where RPN loss $\mathcal{L}_{\text{rpn}}$ is defined as

$$\mathcal{L}_{\text{rpn}} = \mathcal{L}_c^b + \mathcal{L}_r \tag{12}$$

and RCNN loss $\mathcal{L}_{\text{rcnn}}$ is defined as

$$\mathcal{L}_{\text{rcnn}} = \mathcal{L}_c^m + \mathcal{L}_r. \tag{13}$$

Here, the term $\mathcal{L}_c^b$ indicates the binary cross-entropy loss over two categories (foreground and background), and the term $\mathcal{L}_c^m$ indicates the categorical cross-entropy loss over all the categories. The term $\mathcal{L}_r$ represents the box regression loss with standard smooth L1 loss defined in Faster RCNN [13].

*Phase II (Meta Train):* In the meta training stage, we adopt the episodic training strategy, which has shown better generalization ability on novel few-shot data [23]. The episode is a mini-batch randomly sampled from each class in the dataset containing base and novel sets' data. The proposed meta memory module stores the information for each class and is optimized with a momentum updating strategy to capture the discriminative information progressively. Furthermore, for class-agnostic RPN, we adopt the DAFE module to filter out the features not matched to any class. For the class-specific detection head, we adopt the MFE module to enhance the RoI features. The loss function is defined as

$$\mathcal{L}_{\text{MMRCNN}} = \mathcal{L}_{\text{FRCNN}} + \lambda_r \mathcal{L}_r \tag{14}$$

where $\mathcal{L}_{\text{FRCNN}}$ is defined in (11), and $\mathcal{L}_r$ is defined in (6). $\lambda_r$ is a hyperparameter.

*Phase III (Meta Test):* Unlike other meta-learning-based approaches, the meta testing stage of our method is different from the meta training stage. We do not need a support branch (the blue path in Fig. 3) in the meta testing, and meta memory will not be updated. The MM-RCNN detects all the classes with the guidance of meta memory, as shown in Fig. 2.

## V. EXPERIMENTS

In this section, we first introduce the adopted remote sensing benchmarks and evaluation metrics for performance evaluation. Next, we present the experimental settings in detail. Furthermore, we conduct comparisons of our approach with conventional and some recently FSOD approaches to verify the effectiveness. Finally, we analyze the impact of our proposed modules in the ablation experiments.

### A. Datasets and Evaluation Metrics

DIOR [2] is a large remote sensing object detection benchmark which consists of 23 463 images and 192 472 manually labeled objects of 20 classes. The object classes of this dataset include airplane (APL), airport (APO), baseball-field (BF), basketball-court (BC), bridge (BR), chimney (CH), expressway-service-area (ESA), expressway-toll-station (ETS), dam (DAM), golf-field (GF), ground-track-field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage-tank (STO), tennis-court (TC), train-station (TS), vehicle (VE), and windmill (WM).

NWPU VHR-10 [63] is a widely used remote sensing object detection benchmark containing 800 images and 2934 manually annotated instances over ten classes: airplane (APL), baseball-diamond (BD), basketball-court (BC), bridge (BR), ground-track-field (GTF), harbor (HA), ship (SH), storage-tank (ST), tennis-court (TC), and vehicle (VE). There are 150 images without any target objects of given classes and 650 images with at least one target.

As other FSOD methods, mean average precision (mAP) is adopted to measure the performance of object detection. To calculate mAP, we use the Pascal VOC2007 benchmark [11] with an intersection over union (IoU) of 0.5. Each case is repeated four times, with the average value serving as the final detection result.

### B. Experimental Settings

To fairly compare the proposed method with other FSOD approaches, we follow the same split setting on novel and base classes as FSODM [53] for NWPU VHR-10, i.e., APL, BD, and TC are chosen to be novel classes, and the rest are base classes. The images of the NWPU VHR-10 dataset are resized to 1024 × 1024 pixels, and we keep the original ratios. For DIOR, we follow the same division settings as P-CNN [54] and obtain four different splits on novel and base classes, as illustrated in Table I. Since all the images of the DIOR dataset have the same spatial size of 800 × 800 pixels, we do not need to resize them.

The proposed MM-RCNN is implemented based on the PyTorch [64]. We adopt ResNet101 [10] pretrained on ImageNet as the backbone. The images containing both the base classes and novel classes objects are deleted in the data preprocess stage to ensure the proposed method would not

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TABLE II

FEW-SHOT DETECTION RESULTS FOR THE BASE AND NOVEL CLASSES ON THE DIOR DATASET UNDER 3-, 5-, 10-, 20-, AND 30-SHOT SETTINGS AND DIFFERENT SPLITS. THE BEST RESULTS ARE MARKED IN BOLD FONT

| # shots | Method | Class Split 1 | | Class Split 2 | | Class Split 3 | | Class Split 4 | | Average | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Novel Classes | Base Classes | Novel Classes | Base Classes | Novel Classes | Base Classes | Novel Classes | Base Classes | Novel Classes | Base Classes |
| 3 | Faster RCNN [13] | 0 | 0.459 | 0 | 0.457 | 0 | 0.51 | 0 | 0.484 | 0 | 0.478 |
| | YOLOX [66] | 0 | 0.465 | 0 | 0.451 | 0 | 0.521 | 0 | 0.503 | 0 | 0.485 |
| | RepMet [65] | 0.113 | 0.478 | 0.052 | **0.496** | 0.055 | 0.487 | 0.109 | 0.474 | 0.082 | 0.484 |
| | TFA [27] | 0.116 | 0.477 | 0 | 0.488 | 0.111 | 0.519 | 0.094 | 0.490 | 0.080 | 0.494 |
| | FSCE [26] | 0.123 | 0.489 | 0.096 | 0.478 | 0.125 | **0.526** | 0.105 | 0.512 | 0.112 | **0.501** |
| | Meta YOLO [25] | 0.075 | **0.494** | 0.048 | 0.485 | 0.078 | 0.455 | 0.037 | 0.482 | 0.059 | 0.479 |
| | Meta RCNN [24] | 0.140 | 0.463 | 0.105 | 0.482 | 0.122 | 0.495 | 0.116 | 0.504 | 0.121 | 0.486 |
| | P-CNN [54] | 0.180 | 0.470 | 0.145 | 0.489 | 0.165 | 0.495 | 0.152 | 0.498 | 0.161 | 0.488 |
| | MM-RCNN | **0.198** | 0.472 | **0.156** | 0.487 | **0.167** | 0.504 | **0.164** | **0.517** | **0.171** | 0.495 |
| 5 | Faster RCNN [13] | 0 | 0.441 | 0 | 0.463 | 0 | 0.487 | 0 | **0.518** | 0 | 0.477 |
| | YOLOX [66] | 0 | 0.456 | 0 | 0.488 | 0 | 0.513 | 0 | 0.503 | 0 | 0.490 |
| | RepMet [65] | 0.113 | 0.494 | 0.056 | **0.511** | 0.057 | 0.496 | 0.116 | 0.492 | 0.086 | 0.498 |
| | TFA [27] | 0.162 | 0.488 | 0.036 | 0.495 | 0.138 | 0.524 | 0.096 | 0.496 | 0.108 | 0.501 |
| | FSCE [26] | 0.178 | 0.496 | 0.086 | 0.498 | 0.149 | **0.544** | 0.111 | 0.508 | 0.131 | **0.512** |
| | Meta YOLO [25] | 0.121 | **0.497** | 0.070 | 0.468 | 0.137 | 0.479 | 0.068 | 0.485 | 0.099 | 0.482 |
| | Meta RCNN [24] | 0.168 | 0.482 | 0.125 | 0.502 | 0.154 | 0.500 | 0.136 | 0.512 | 0.146 | 0.499 |
| | P-CNN [54] | 0.228 | 0.484 | 0.149 | 0.491 | 0.188 | 0.499 | 0.175 | 0.499 | 0.185 | 0.493 |
| | MM-RCNN | **0.239** | 0.488 | **0.155** | 0.499 | **0.197** | 0.495 | **0.187** | 0.507 | **0.194** | 0.497 |
| 10 | Faster RCNN [13] | 0 | 0.442 | 0.018 | 0.432 | 0 | 0.491 | 0 | 0.508 | 0.004 | 0.468 |
| | YOLOX [66] | 0 | 0.481 | 0 | 0.501 | 0 | 0.512 | 0 | 0.510 | 0 | 0.501 |
| | RepMet [65] | 0.121 | 0.493 | 0.059 | 0.512 | 0.074 | 0.523 | 0.128 | 0.505 | 0.095 | 0.508 |
| | TFA [27] | 0.229 | 0.492 | 0.052 | 0.485 | 0.146 | **0.528** | 0.114 | 0.485 | 0.135 | 0.498 |
| | FSCE [26] | 0.239 | 0.504 | 0.084 | 0.506 | 0.171 | 0.517 | 0.136 | 0.497 | 0.158 | 0.506 |
| | Meta YOLO [25] | 0.181 | 0.495 | 0.090 | 0.464 | 0.138 | 0.445 | 0.072 | 0.457 | 0.120 | 0.465 |
| | Meta RCNN [24] | 0.220 | 0.494 | 0.153 | 0.503 | 0.203 | 0.522 | 0.170 | **0.523** | 0.186 | 0.511 |
| | P-CNN [54] | 0.276 | **0.509** | 0.189 | **0.525** | 0.234 | 0.521 | 0.189 | 0.517 | 0.222 | **0.518** |
| | MM-RCNN | **0.288** | 0.501 | **0.201** | 0.521 | **0.250** | 0.524 | **0.203** | 0.521 | **0.235** | 0.517 |
| 20 | Faster RCNN [13] | 0.071 | 0.458 | 0.018 | 0.463 | 0.112 | 0.485 | 0.024 | 0.452 | 0.056 | 0.465 |
| | YOLOX [66] | 0.081 | 0.471 | 0.035 | 0.478 | 0.157 | 0.509 | 0.081 | 0.511 | 0.089 | 0.492 |
| | RepMet [65] | 0.122 | 0.486 | 0.068 | 0.504 | 0.078 | 0.524 | 0.167 | 0.535 | 0.109 | 0.512 |
| | TFA [27] | 0.237 | 0.491 | 0.078 | 0.485 | 0.189 | 0.524 | 0.128 | 0.502 | 0.158 | 0.501 |
| | FSCE [26] | 0.245 | 0.509 | 0.089 | 0.493 | 0.204 | 0.541 | 0.142 | 0.515 | 0.170 | 0.515 |
| | Meta YOLO [25] | 0.220 | 0.500 | 0.141 | 0.435 | 0.185 | 0.435 | 0.122 | 0.444 | 0.167 | 0.454 |
| | Meta RCNN [24] | 0.243 | 0.493 | 0.205 | 0.523 | 0.263 | 0.537 | 0.220 | **0.534** | 0.233 | 0.522 |
| | P-CNN [54] | 0.296 | 0.522 | 0.228 | 0.516 | 0.288 | 0.531 | 0.257 | 0.523 | 0.267 | 0.523 |
| | MM-RCNN | **0.308** | **0.529** | **0.239** | **0.525** | **0.305** | **0.536** | **0.271** | 0.531 | **0.281** | **0.530** |
| 30 | Faster RCNN [13] | 0.198 | 0.444 | 0.053 | 0.443 | 0.177 | 0.443 | 0.079 | 0.492 | 0.127 | 0.456 |
| | YOLOX [66] | 0.204 | 0.465 | 0.082 | 0.457 | 0.195 | 0.487 | 0.114 | 0.501 | 0.149 | 0.478 |
| | RepMet [65] | 0.151 | 0.509 | 0.065 | 0.507 | 0.086 | 0.507 | 0.173 | 0.512 | 0.119 | 0.509 |
| | TFA [27] | 0.240 | 0.506 | 0.098 | 0.492 | 0.199 | 0.526 | 0.145 | 0.505 | 0.170 | 0.507 |
| | FSCE [26] | 0.261 | 0.515 | 0.125 | 0.515 | 0.213 | 0.523 | 0.161 | 0.521 | 0.190 | 0.519 |
| | Meta YOLO [25] | 0.296 | 0.504 | 0.144 | 0.444 | 0.231 | 0.452 | 0.159 | 0.417 | 0.207 | 0.454 |
| | Meta RCNN [24] | 0.281 | 0.518 | 0.235 | 0.532 | 0.290 | 0.546 | 0.253 | 0.537 | 0.264 | 0.533 |
| | P-CNN [54] | 0.326 | 0.533 | 0.257 | **0.530** | 0.321 | **0.536** | 0.287 | 0.536 | 0.298 | 0.534 |
| | MM-RCNN | **0.341** | **0.541** | **0.272** | 0.527 | **0.345** | 0.534 | **0.302** | **0.538** | **0.315** | **0.535** |

see the novel classes' object in the base training stage. SGD is adopted as the optimizer, the base learning rate is set to 0.001, and the momentum coefficient and weight decay are set to 0.9 and 0.0001, respectively. We train our MM-RCNN for 20 epochs for the base training phase and ten epochs for the meta training phase. We used one Nvidia Titan XP GPU to accelerate the training and testing in all the experiments.

### C. Experimental Results and Comparisons

*1) Results on DIOR Dataset:* We compared the proposed MM-RCNN with several FSOD methods, including the fine-tuning-based methods (RepMet [65], TFA [27] and FSCE [26]) and meta-learning-based methods (MetaYOLO [25], Meta RCNN [24], and P-CNN [54]), as well as the conventional object detector Faster RCNN [13] and YOLOX [66]. The results of all the cases are listed in Table II.

We draw the following observations from Table II. First, the conventional object detector Faster R-CNN and YOLOX perform poorly without sufficient training samples, especially in low-shot cases, e.g., $K = 3, 5, 10$. The main reason is that the conventional object detectors are trained on a balanced dataset and prone to overfit on the few-shot scenario. Second, the fine-tuning-based few-shot detectors perform better than Faster RCNN and YOLOX, demonstrating that the fine-tuning strategy is simple and effective in the overfitting problem caused by imbalanced training data. Third, compared with the fine-tuning-based FSOD approaches, meta-learning-based detectors achieve better results, which proves that the meta-learning strategy can enable fast adaptation to novel classes. For example, P-CNN achieved an average mAP of 0.161, 0.185, 0.222, 00267, and 0.298 across different shots. Finally, the proposed MM-RCNN is a meta-learning-based detector and surpasses the other meta-learning-based detectors in most cases. Compared with Meta RCNN and P-CNN, which take the averaged support samples as the class-specific prototype to reweight the features, the proposed MM-RCNN introduces a memory mechanism that gradually learns a more discriminative representation for each class. As shown in Table II, the proposed MM-RCNN model obtains an average mAP of 0.171,

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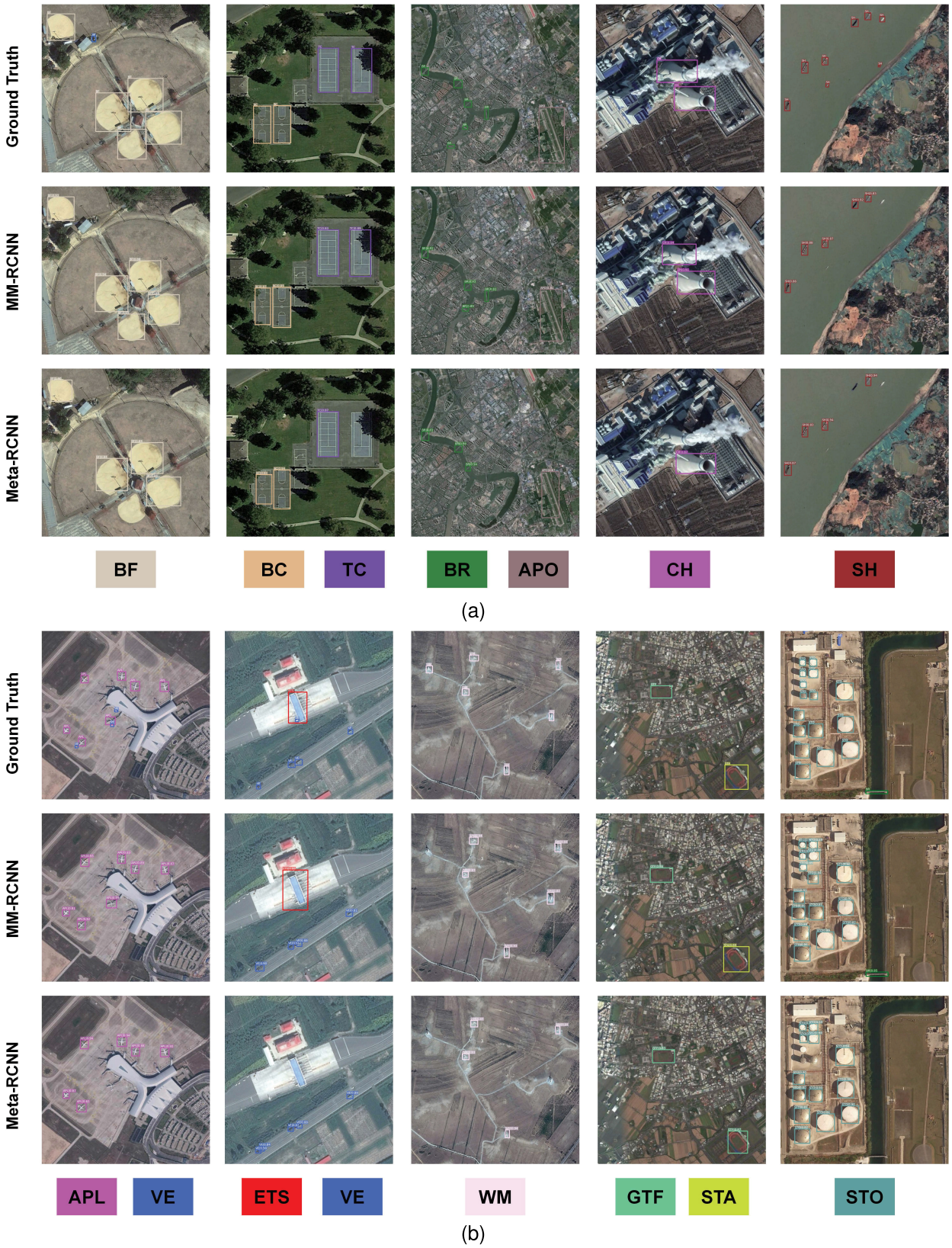



Fig. 8. Examples of the detection results in our MM-RCNN and Meta-RCNN for novel and base classes under split1 and 30-shot settings on the DIOR dataset. (a) Detection results for novel classes. (b) Detection results for base classes. Different categories are marked with different colors.

0.194, 0.235, 0.281, and 0.315 among different shots on novel classes. MM-RCNN achieves the mean mAP of 0.239 across all the settings, which outperforms the second-best approach P-CNN.

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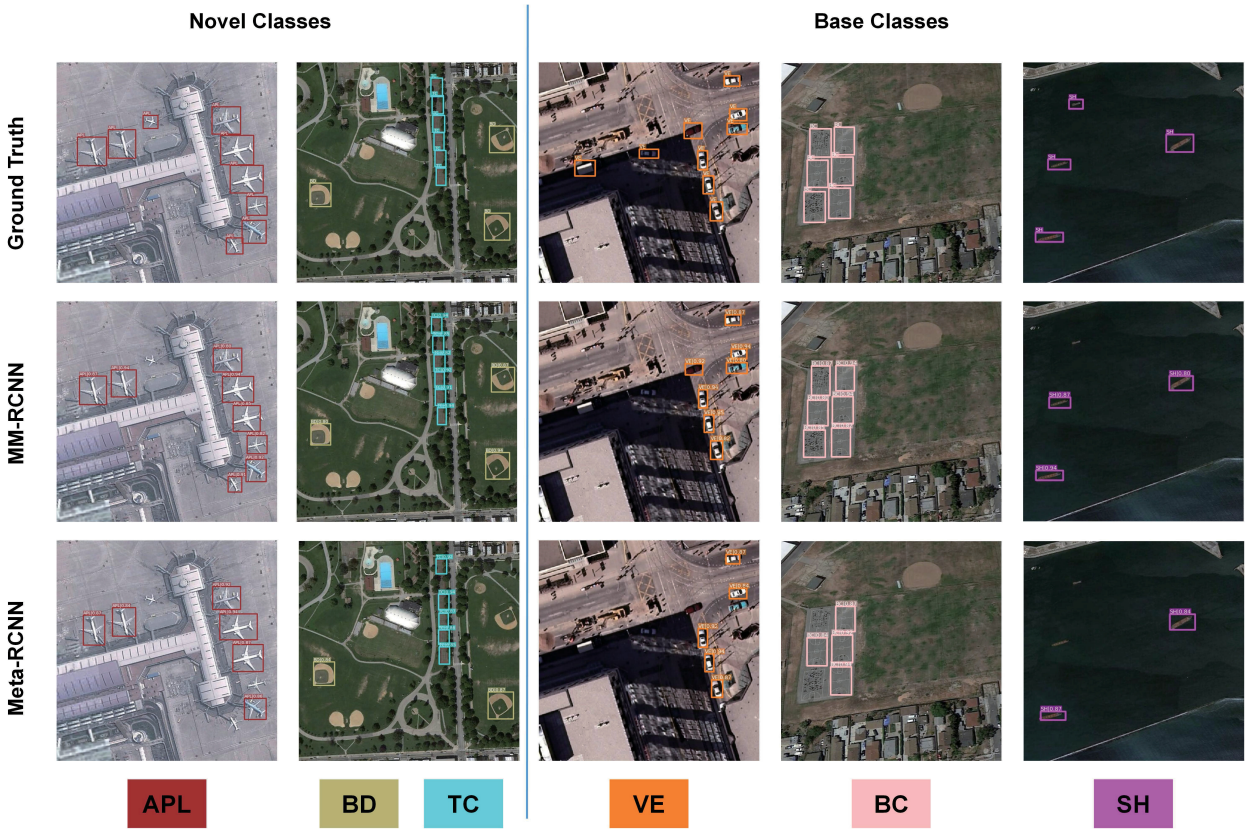



Fig. 9. Examples of the detection results in our MM-RCNN and Meta-RCNN for novel and base classes under 10-shot setting on the NWPU-VHR10 dataset. Different categories are marked with different colors.

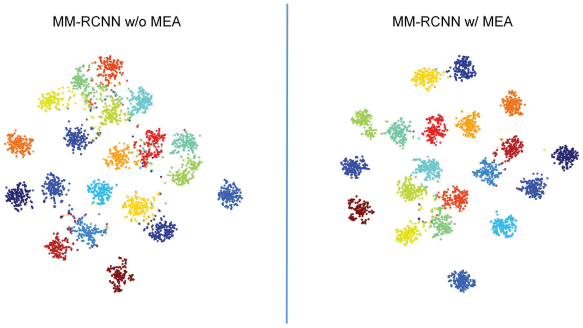



Fig. 10. $t$-SNE visualization of the embedding space without and with the proposed MEA. Different colors indicate different classes.

Some detection results are shown in Fig. 8. We depict the ground truth in the first row and the detection results of the proposed MM-RCNN and Meta-RCNN [24] in the second and third rows. We can see from the second row that our method can detect the majority of objects that are either base classes or novel classes, demonstrating the efficacy of the proposed method. Furthermore, we observe some failure cases, which are primarily caused by the complex background and large orientation and size variations. The research of how to extract robust scale- and rotation-invariant features in FSOD will be the focus of our future study.

*2) Results on NWPU VHR-10 Dataset:* We compared our MM-RCNN with some FSOD methods, including the fine-tuning-based methods (TFA [27], FSCE [26], and PAMSDet [55]) and meta-learning-based methods (MetaYOLO [25], Meta RCNN [24], and FSODM [53]), as well as the conventional object detector Faster RCNN [13] and YOLOX [66]. All the results are listed in Table III.

We observe that the proposed MM-RCNN surpasses all the competing methods in most cases and achieves the best average mAP from Table III. The conventional object detectors Faster RCNN and YOLOX still obtain a poor performance than all the few-shot detectors. Even though Faster RCNN obtains an mAP of 0.38 in 10-shot, the performance is still worse than most few-shot methods in 3-shot. PAMSDet is a fine-tuning-based few-shot detector and achieves the state-of-the-art performance of 0.37, 0.55, and 0.66 mAP on novel classes in three different settings. Finally, the proposed MM-RCNN achieves an average mAP of 0.557 among three settings. Specifically, our approach obtains an mAP of 0.43, 0.56, and 0.68 on novel classes in three different settings. From Table III, we observe that our approach achieves mAP 6% higher, 1% higher, and 2% higher than PAMSDet in 3-shot, 5-shot, and 10-shot, respectively. Some detection examples are depicted in Fig. 9.

### *D. Ablation Studies*

We conducted extensive ablation studies on different shot and four different base/novel split settings on the DIOR dataset to evaluate the importance of the proposed meta memory module, the DAFE module, and the MFE module. The results of all the cases are listed in Tables IV–VI.

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TABLE III

FEW-SHOT DETECTION RESULTS FOR THE NOVEL CLASSES ON THE NWPU VHR-10 DATASET UNDER 3-, 5-, AND 10-SHOT SETTINGS. THE BEST RESULTS ARE MARKED IN BOLD FONT

| # shots | Method | APL | BD | TC | Mean |
|---|---|---|---|---|---|
| 3 | Faster RCNN [13] | 0 | 0 | 0 | 0 |
| | YOLOX [67] | 0.03 | 0 | 0 | 0.01 |
| | TFA [27] | 0.12 | 0.61 | 0.13 | 0.29 |
| | FSCE [26] | 0.13 | 0.58 | 0.11 | 0.27 |
| | PAMSDet [56] | 0.21 | 0.76 | 0.16 | 0.37 |
| | Meta YOLO [25] | 0.13 | 0.12 | 0.11 | 0.12 |
| | FSODM [54] | 0.15 | 0.57 | **0.25** | 0.32 |
| | Meta RCNN [24] | 0.18 | 0.61 | 0.20 | 0.33 |
| | MM-RCNN | **0.23** | **0.81** | 0.24 | **0.43** |
| 5 | Faster RCNN [13] | 0.11 | 0.24 | 0.06 | 0.14 |
| | YOLOX [67] | 0.15 | 0.21 | 0.09 | 0.15 |
| | TFA [27] | 0.51 | 0.78 | 0.19 | 0.49 |
| | FSCE [26] | 0.48 | 0.76 | 0.16 | 0.47 |
| | PAMSDet [56] | 0.55 | 0.88 | 0.20 | 0.55 |
| | Meta YOLO [25] | 0.24 | 0.39 | 0.11 | 0.24 |
| | FSODM [54] | **0.58** | 0.84 | 0.16 | 0.53 |
| | Meta RCNN [24] | 0.50 | 0.83 | 0.17 | 0.50 |
| | MM-RCNN | 0.57 | **0.89** | **0.21** | **0.56** |
| 10 | Faster RCNN [13] | 0.34 | 0.58 | 0.21 | 0.38 |
| | YOLOX [67] | 0.36 | 0.61 | 0.25 | 0.41 |
| | TFA [27] | 0.60 | 0.85 | 0.49 | 0.65 |
| | FSCE [26] | 0.57 | 0.82 | 0.46 | 0.62 |
| | PAMSDet [56] | 0.61 | 0.88 | 0.50 | 0.66 |
| | Meta YOLO [25] | 0.20 | 0.74 | 0.26 | 0.40 |
| | FSODM [54] | 0.60 | 0.88 | 0.48 | 0.65 |
| | Meta RCNN [24] | 0.50 | 0.83 | 0.45 | 0.59 |
| | MM-RCNN | **0.63** | **0.90** | **0.51** | **0.68** |

TABLE IV

ABLATION STUDIES ON THE DIOR DATASET UNDER DIFFERENT SPLITS. THE MAP IS ADOPTED AS EVALUATION METRIC. THE BEST RESULTS ARE MARKED IN BOLD FONT

| # shots | DAFE | MFE | Split 1 | Split 2 | Split 3 | Split 4 | Average |
|---|---|---|---|---|---|---|---|
| 3 | ✓ | | 0.125 | 0.099 | 0.108 | 0.104 | 0.109 |
| | | ✓ | 0.145 | 0.114 | 0.132 | 0.115 | 0.127 |
| | ✓ | ✓ | **0.198** | **0.156** | **0.167** | **0.164** | **0.171** |
| 5 | ✓ | | 0.192 | 0.124 | 0.175 | 0.157 | 0.162 |
| | | ✓ | 0.207 | 0.141 | 0.182 | 0.172 | 0.176 |
| | ✓ | ✓ | **0.239** | **0.155** | **0.197** | **0.187** | **0.195** |
| 10 | ✓ | | 0.227 | 0.168 | 0.195 | 0.167 | 0.189 |
| | | ✓ | 0.264 | 0.183 | 0.214 | 0.179 | 0.210 |
| | ✓ | ✓ | **0.288** | **0.201** | **0.250** | **0.203** | **0.236** |
| 20 | ✓ | | 0.271 | 0.197 | 0.268 | 0.225 | 0.240 |
| | | ✓ | 0.284 | 0.214 | 0.281 | 0.234 | 0.253 |
| | ✓ | ✓ | **0.308** | **0.239** | **0.305** | **0.271** | **0.281** |
| 30 | ✓ | | 0.304 | 0.251 | 0.297 | 0.267 | 0.280 |
| | | ✓ | 0.321 | 0.258 | 0.312 | 0.284 | 0.294 |
| | ✓ | ✓ | **0.341** | **0.272** | **0.345** | **0.302** | **0.315** |

*1) Impact of the DAFE Module:* The DAFE module is designed to aggregate the information of all the classes simultaneously to capture the interclass correlation and filter out the features not matched to any category for the purpose of reducing the false classification of RPN. The results in Table IV proved that the method with the DAFE module performs better, like 10-shot (+2.6% mAP) and 20-shot (+2.8% mAP).

*2) Impact of the MFE Module:* Table IV also validates the effectiveness of the MFE module, which calculates the soft attention between the meta memory and RoI features to improve invariance and retain the information of the original RoI features. We observe that the method with the MFE module performs better, like 20-shot (+4.1% mAP) and 30-shot (+3.5% mAP).

TABLE V

ABLATION STUDIES OF DIFFERENT EXTERNAL GUIDANCE USED IN MM-RCNN. "P" INDICATES USING PROTOTYPES AS GUIDANCE, AND "MM" INDICATES USING META MEMORY AS GUIDANCE. THE MAP IS ADOPTED AS EVALUATION METRIC. THE BEST RESULTS ARE MARKED IN BOLD FONT

| # shots | MM | P | Split 1 | Split 2 | Split 3 | Split 4 | Average |
|---|---|---|---|---|---|---|---|
| 3 | | ✓ | 0.181 | 0.142 | 0.148 | 0.151 | 0.156 |
| | ✓ | | **0.198** | **0.156** | **0.167** | **0.164** | **0.171** |
| 5 | | ✓ | 0.197 | 0.148 | 0.183 | 0.169 | 0.174 |
| | ✓ | | **0.239** | **0.155** | **0.197** | **0.187** | **0.195** |
| 10 | | ✓ | 0.243 | 0.168 | 0.208 | 0.175 | 0.199 |
| | ✓ | | **0.288** | **0.201** | **0.250** | **0.203** | **0.236** |
| 20 | | ✓ | 0.254 | 0.203 | 0.275 | 0.224 | 0.239 |
| | ✓ | | **0.308** | **0.239** | **0.305** | **0.271** | **0.281** |
| 30 | | ✓ | 0.285 | 0.236 | 0.302 | 0.264 | 0.272 |
| | ✓ | | **0.341** | **0.272** | **0.345** | **0.302** | **0.315** |

TABLE VI

ABLATIONS STUDIES OF MEA USED IN MM-RCNN. "MEA" INDICATES THE PROPOSED MEA, AND "FR" INDICATES FEATURE REWEIGHTING. THE MAP IS ADOPTED AS EVALUATION METRIC. THE BEST RESULTS ARE MARKED IN BOLD FONT

| # shots | MEA | FR | Split 1 | Split 2 | Split 3 | Split 4 | Average |
|---|---|---|---|---|---|---|---|
| 3 | | ✓ | 0.175 | 0.137 | 0.141 | 0.153 | 0.152 |
| | ✓ | | **0.198** | **0.156** | **0.167** | **0.164** | **0.171** |
| 5 | | ✓ | 0.186 | 0.134 | 0.175 | 0.165 | 0.165 |
| | ✓ | | **0.239** | **0.155** | **0.197** | **0.187** | **0.195** |
| 10 | | ✓ | 0.231 | 0.149 | 0.187 | 0.164 | 0.183 |
| | ✓ | | **0.288** | **0.201** | **0.250** | **0.203** | **0.236** |
| 20 | | ✓ | 0.231 | 0.192 | 0.252 | 0.201 | 0.220 |
| | ✓ | | **0.308** | **0.239** | **0.305** | **0.271** | **0.281** |
| 30 | | ✓ | 0.258 | 0.219 | 0.285 | 0.249 | 0.253 |
| | ✓ | | **0.341** | **0.272** | **0.345** | **0.302** | **0.315** |

*3) Impact of the Meta Memory Module:* The prevalent meta-learning-based few-shot object detectors [24], [54] adopt prototypes representing the cluster centroid as guidance to reweight the features. In contrast, we propose a meta memory module to gradually learn the more discriminative information of the class through episodic training. From Table V, we observe that using the meta memory brings considerable improvement over prototypes. The performance gain is negligible under extremely low-shots like 3-shot (+1.5% mAP) and 5-shot (+2.1% mAP), and the performance gain is significant under high-shots, particularly under 20-shot (+4.2% mAP) and 30-shot (+4.3% mAP).

*4) Impact of the MEA:* MEA is proposed to capture the interclass correlations, which is important for the category classification. From Table VI, we observe that the proposed MEA achieves better average mAP among different shots, especially under 30-shot (+6.2% mAP) and 20-shot (+6.1% mAP), which verifies the effectiveness of the interclass correlations. Moreover, Fig. 10 shows t-SNE [67] visualization of the embedding space with the objects from different classes without and with the proposed MEA. The results show that the embedding space of MM-RCNN without MEA is mixed, which may cause misclassification. On the contrary, with the proposed MEA to aggregate all the classes simultaneously,

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different classes in the embedding space have large margins with each other, which verifies the effectiveness of the proposed MEA module.

## VI. Conclusion

In this article, we propose a novel meta-memory-based approach named MM-RCNN for FSOD task in remote sensing scenes. Specifically, we adopt the memory mechanism to store each class' meta knowledge, gradually aggregating discriminative information and diluting object-irrelevant nuisances in the meta training stage. Then we propose the MEA module to aggregate all the categories simultaneously and capture the interclass correlation to reduce false classification. Finally, we propose two feature enhancement modules, i.e., DAFE for RPN and MFE for detection head, to boost the performance of the detector. Experiments on two publicly available remote sensing object detection benchmarks, DIOR and NWPU VHR-10, prove that our approach is capable of detecting objects from both the base and novel classes under the few-shot scenario.

Nevertheless, the objects of interest in remote sensing images always have large variations on scales, aspect ratios, and orientations, which hinders the detection performance when the samples are insufficient. In our future study, we will concentrate on the study of the rotation- and scale-invariant feature extraction using a few training examples.

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